CONFIDENTIAL — DRAFT — FOR DISCUSSION · SPECIMEN AND CONDITIONING ARE INVENTOR-DEFINED AND TAGGED

Characterize first. Condition second. Claim never, until the accounting closes. The plan for the first named research module.

The Oct-13 deck names Floyd Sweet's solid-state design — the vacuum triode amplifier — as the first research module, to be characterized, tested against the energy-conservation filter and then extracted or retired. This plan turns that slide into an experiment that can be pre-registered. It follows the baseline protocol in the Physical Architecture & Baseline Model v1.3 and the artifact-exclusion discipline of the September prospectus. It contains no performance claim and no device. Every input only the inventor can supply is tagged.

1 · WHAT IS BEING TESTED, STATED SO IT CAN BE FALSIFIED

> **First testable question.** Does a defined conditioning procedure produce a repeatable change in magnetic response or in loaded conversion performance of a core-and-coil assembly, compared with a matched control specimen that received a sham treatment? A measured change is engineering. A claim of additional energy requires full closure of the energy balance: $E_{\text{all inputs}} = E_{\text{load}} + E_{\text{other outputs}} + \Delta U_{\text{device}}$.

Three physical hypotheses are carried into the plan from the baseline model, each with the observation that would support or retire it:

Hypothesis	Physical statement	Discriminating observation
H-A1 · Finite stored-energy depletion	Conditioning leaves the core in a non-equilibrium domain, exchange-bias or strain state; excitation triggers relaxation that releases finite stored energy.	Output declines with cumulative delivered energy; remanence and coercivity change in step; the integrated surplus is bounded by a measured change in core state. Hysteresis loss is integrated over the core volume, not scaled from a single B–H loop.
H-A2 · Open-system transduction	The assembly harvests ambient energy — thermal gradient, RF flux or mechanical vibration.	Output tracks the ambient term when it is varied or shielded: ΔT sweeps and calorimetry; characterized RF enclosure on and off; vibration isolation on and off.
H-A3 · Metrology phase skew	Probe timing deskew in a high-Q circuit distorts the computed real power; an apparent surplus is an instrument artifact.	Reading changes with frequency as the deskew model predicts; disappears when the whole measurement chain is calibrated and when power is computed from $\int v(t)i(t)dt$ rather than $V_{RMS} \cdot I_{RMS} \cdot \cos\theta$.

2 · THE SPECIMEN — INVENTOR-DEFINED ITEMS

Item	Baseline proposal (v1.3, unvalidated)	Input required
Core material and grade	Sintered strontium ferrite ($SrFe_{12}O_{19}$), Grade 8 permanent magnet — tests the Sweet lineage; controls matched in material, geometry and preparation; soft ferrite only as an instrumentation reference.	**[Needs Input from Ian or Chip: the actual core — material, grade, supplier, lot; or approval of the off-the-shelf baseline]**
Geometry and magnetization	Toroid OD 75 mm, ID 50 mm, H 20 mm; or block 150 × 100 × 25 mm. Magnetization axis and initial state recorded before anything is wound.	**[Needs Input from Ian or Chip: geometry, magnetization direction and state, winding axis relative to magnetization]**
Excitation winding N1	50 turns, AWG 20 enamelled copper — proposed; resistance, AC loss and thermal limit validated against measured current and duty.	Confirm or replace **[Needs Input from Ian or Chip: winding specification if the original design differs]**
Pickup winding N2	200 turns, AWG 24 — proposed; the step-up gives a measurable loaded voltage across the test resistor.	Confirm or replace
Load	50 Ω non-inductive power resistor, Kelvin-sensed; impedance and temperature dependence measured as a component.	—
Drive	Linear RF power amplifier from a synthesized generator; sine, THD < 0.1 %, 10–100 kHz, within the specimen's characterized range.	—
Conditioning procedure	None selected. No capacitor-discharge, saturation-soak or mechanical protocol has been chosen; the baseline is run unconditioned first.	**[Needs Input from Ian or Chip: the intended physical state change, the procedure, and the energy delivered during conditioning]**
Historical record	The design's documented conversion chain — resonant LC tank, rectifier / charge pump, DC-link buffer, feedback controller — is standard conversion engineering; the mechanism supplying energy above the conventional baseline is the unestablished hypothesis.	**[Needs Input from Ian or Chip: the founder's notes, drawings and any prior measurements on the module, for the pre-registration appendix]**

3 · THE FOUR-STEP PROTOCOL

Step	What is done	Rule
1 · One specimen	Select one concrete hardware specimen and record material, geometry, magnetic orientation and initial state. Photograph, weigh, measure. Prepare matched control specimens from the same lot.	No measurement before the specimen record is frozen.
2 · Characterize before conditioning	Winding DC and AC resistance; impedance versus frequency; coil coupling; temperature-dependent magnetic response; remanence and coercivity proxies on the control pair.	Characterization drives can alter the core's state: record every drive level; monitor initial and final states.

Step	What is done	Rule
3 · Externally driven, loaded	Excite within the characterized range; deliver power to the fixed resistive load; integrate input and output simultaneously: $E_{exc} = \int v_{exc} i_{exc} dt$, $E_{load} = \int i_{load}^2 R\, dt$. Map efficiency $\eta = P_{load}/P_{input}$ across frequency and drive.	DC input is ∫v·i, never average V × average I. The deskew check (§4) runs at every frequency used.
4 · Conditioning as a separate experiment	Only after the unconditioned map is complete. Define the intended state change; meter total conditioning energy $E_{cond} = \int v_{cond} i_{cond} dt$; apply a sham treatment to the control; repeat Steps 2–3 on both, blinded.	Pre-registered before the first conditioning run; the comparison is conditioned versus sham, not before versus after.

4 · ENERGY ACCOUNTING AND THE METROLOGY CHECK

> **$E_{\text{all inputs}} = E_{\text{load}} + E_{\text{heat and other outputs}} + \Delta U_{\text{device}}$**, where U includes energy stored in capacitors, batteries, magnetic and strain states and internal thermal reservoirs. For steady operation (ΔU = 0) the efficiency is $\eta = P_{load}/P_{\text{total external input}}$; internal losses are already inside the input and are not added to the denominator. Closed-loop feedback with forward efficiency $\eta_{fwd} < 1$ and feedback fraction $\beta \le 1$ has loop gain $T = \beta \cdot \eta_{fwd} < 1$ and decays geometrically; self-sustained operation without an independent input or stored-energy depletion does not occur.

Deskew benchmark (v1.3 §3, hypothetical): 10 VA apparent power, +50 mW true dissipation, 2 ns probe timing error — the computed real power reads +49.87 mW at 1 kHz, +37.43 mW at 100 kHz, 0 at ≈398 kHz and −75.66 mW at 1 MHz: a false “generation” reading. The plan therefore calibrates the complete measurement chain, computes power from sampled v(t)i(t), and repeats the loaded map with the probe pair swapped.

5 · ARTIFACT CATALOGUE FOR A CORE-AND-COIL SYSTEM

Artifact source	Mechanism	Discriminating test	Signature if artifact
Probe phase skew	Timing error between voltage and current probes in a reactive circuit.	Frequency sweep against the deskew model; probe swap; sampled-waveform integration.	Apparent surplus grows with frequency; vanishes under full-chain calibration.
Stored magnetic energy	Conditioning stores energy in the core state; later excitation releases it.	Track cumulative delivered energy; remanence and coercivity before and after; volume-integrated hysteresis loss.	Output decays with delivered energy; core state changes in step.
Thermal transduction	A temperature difference across the assembly drives a thermoelectric or thermomagnetic contribution.	Mapped ΔT; deliberate ±ΔT sweeps; calorimetric heat-flow measurement.	Output tracks ΔT and vanishes as ΔT → 0.
Ambient electromagnetic flux	Windings and core couple to ambient RF; the rectifier converts it to DC.	Shielded enclosure with characterized attenuation; injected-RF response; spectrum monitoring.	Output correlates with ambient RF and shielding state. An illustrative 1 μW/cm² over 150 cm² is 150 μW incident — absorbed power requires measured coupling.

Artifact source	Mechanism	Discriminating test	Signature if artifact
Capacitive and battery storage	DC-link or buffer elements supply energy stored earlier.	Full discharge before runs; meter every storage element; account ΔU explicitly.	Output bounded by stored energy; decays with the storage time constant.
Instrument offsets and bias	Meter offsets and bias currents read as output.	Open-circuit, shorted and dummy-load runs interleaved.	Same reading with the specimen removed.
Selection and analysis bias	Choosing the “good” runs after the fact.	Pre-registered sequence; blinded conditioned / sham labels; fixed stopping rules; all runs reported.	Effect shrinks in the full population.

6 · PRE-REGISTERED DECISION RULE

Outcome	Definition	Consequence
Pass	A repeatable, conditioning-dependent change in loaded conversion or magnetic response, at ≥5σ against the empirical distribution of sham-treated controls, in blinded runs, surviving every catalogue row, in the full pre-registered population — *and* a closed energy balance that states exactly where any surplus came from.	Report and data published; the module proceeds to Gate 2 conservation testing (closed-system calorimetry, second laboratory). No performance figure is stated.
Kill	No change above the floor; or a change fully explained by one or more catalogue rows; or a surplus accounted for by stored energy or an ambient input.	Published as a null or as a transducer result; the module is retired publicly, as the deck promises.
Ambiguous	A change above the floor that does not survive one row cleanly, or does not reach 5σ in the full population.	Reported as such; one bounded, pre-registered extension; a second ambiguous outcome is a kill.

7 · DELIVERABLES, IN ORDER

1. Specimen record and control-pair record (Step 1), signed by the inventor and the measurement lead.
2. Pre-characterization report (Step 2) with drive levels and initial / final states.
3. Unconditioned loaded-conversion map (Step 3) with the deskew check at every frequency.
4. Conditioning pre-registration: intended state change, procedure, energy metering, sham protocol, blinding plan, stopping rules.
5. Conditioned-versus-sham report under the decision rule above, with raw data timestamped and hashed at acquisition.

8 · INPUTS THIS PLAN CANNOT SUPPLY

- **[Needs Input from Ian or Chip: specimen — material, grade, geometry, magnetization, winding axis]**
- **[Needs Input from Ian or Chip: the conditioning procedure and its intended physical state change]**
- **[Needs Input from Ian or Chip: the founder's notes and any prior measurements on the module]**
- **[Needs Input from Ian or Chip: whether the deck's Sweet module and the website's "mid-1980s solid-state design" are the same device]**
- **[Needs Input from Ian or Chip: the measurement lead who signs the specimen record, and the laboratory]**
- **[Needs Input from Ian or Chip: whether thread A or thread B runs first on the shared platform — decision D2]**

Sources: Oct-13 deck slide 5 (S-01); Physical Architecture & Baseline Model v1.3 §§1–5, 8 (M-14 / S-11); Claude physics review v2 §3 (S-13); Technical Prospectus v1 §§5, 6.4–6.6 (M-05); Research Program and Gates page; DECISIONS.md D2.